

Mahmudul Hasan

Software Engineer

mkmahmud.dev@gmail.com | +880 17415-03127
Dhaka | linkedin.com/in/mkmahmud | github.com/mkmahmud

Summary

Full Stack Software Engineer with over two years of professional experience building and shipping production web applications end-to-end. Specialised in TypeScript, React/Next.js, Python/FastAPI, PostgreSQL, and AWS, with hands-on experience building scalable APIs, real-time features, AI-powered applications, RAG systems, and automated workflows across EdTech, healthcare, travel, and e-commerce.

Experience

Full Stack Developer, Codexmine - Dhaka Mar 2026- Present

- ❖ Built full EdTech ERP covering academics, HR, students, attendance, expenses and others end-to-end.
- ❖ Engineered biometric attendance pipeline that eliminated all manual roll-call entry entirely.
- ❖ Automated result engine computes and publishes grades directly from raw exam marks on demand.
- ❖ Computer-vision OMR checker grades bulk-scanned answer sheets and posts scores automatically.
- ❖ Built a RAG chatbot with LangChain and vector search, grounding responses in institutional documents.
- ❖ AI Sales Pilot agent handles lead research, email drafting and outbound sending autonomously.
- ❖ Python and FastAPI AI services deployed on PostgreSQL and AWS RDS with full pytest coverage.

Full Stack Developer, LeoPhoenix, US(Remote) Oct 2023- Feb 2026

- ❖ Built and scaled production application using Next.Js & TypeScript with performance score above 90.
- ❖ Architected full-stack features end-to-end across React, NestJS, PostgreSQL, and Prisma ORM.
- ❖ Designed relational and NoSQL databases, utilizing ORMs and Redis layers to slash API response times.
- ❖ Built responsive production UIs using Tailwind css and Shadcn.
- ❖ Shipped complete Stripe payment flows and OpenAI-powered AI features into production safely.
- ❖ Implemented JWT authentication and RBAC for secure role-based access control.
- ❖ Containerised services with Docker; automated CI/CD to AWS and Vercel for zero-downtime deploys.
- ❖ Leveraged firebase for rapid prototyping and faster feature release.
- ❖ Maintained Jest unit test coverage across all core application flows throughout development.

Skills

Languages: TypeScript, JavaScript (ES6+), Python, SQL, HTML5, CSS3.
Frontend & UI: React.js, Next.js, Redux, Zustand, TanStack Query, Vite, Tailwind CSS, Shadcn UI, GSAP.
Backend: Node.js, NestJS, Express.js, FastAPI, REST APIs, GraphQL, Socket.IO.
Databases: PostgreSQL, MongoDB, MySQL, Redis, Prisma ORM, Mongoose, Firebase.
DevOps & Tools: Linux, Docker, AWS, Vercel, CI/CD, Git, JWT, RBAC, Zod, Jest, Pytest, Swagger, OAuth.
AI Engineering: LangChain, LangGraph, RAG, vector databases, OpenAI API, n8n automation.

Project

Edumine ~ Educational Institute Management System

- Built full-stack institute ERP spanning academics, attendance, exams, results, payroll, and assets.
- Engineered OpenCV OMR engine auto-grading scanned MCQ sheets with alignment fallback, confidence scoring, and review.
- Designed RBAC across 7 roles and 76 permissions with audit logs and result publishing.
- Delivered biometric attendance, conflict-free routines, invoicing, dues tracking, and SMS alerts.

Technology: Next.js, FastAPI, PostgreSQL, SQLAlchemy, OpenCV, Cloudinary, AWS EC2, Nginx

Education

Bachelor's degree, Islamic Studies, Uttara University May, 2022 – Apr, 2026

Certification

Next Level Web Development, Programming Hero 2024
AWS Basics, KodeKloud 2026